

Bell Nozzle Rocket Engine Calculations

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Nomenclature

C_f	Vacuum thrust coefficient
c_s	Speed of sound
$\dot{m}$	Mass flow rate
M	Molar mass
Ma	Mach number
$mol\%$	Mole percentage
P	Pressure
R	Gas constant
R_s	Specific gas constant
T	Temperature
v	Velocity
ϵ	Nozzle area ratio
γ	Ratio of specific heats
ρ	Density

Subscripts

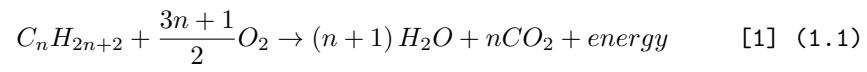
a	Ambient
c	Chamber
e	Exit
s	Stagnation
t	Throat
x	Current Position

Introduction

Documentation of calculations for a bell nozzle rocket engine

1 Propellant

1.1 Propellant Stoichiometry of Hydrocarbon-Oxygen Fuel Mixes



1.2 Mass Flow Rate

$$\dot{m} = \frac{m}{t} \quad [2] \quad (1.2)$$

2 Exhaust Gasses

2.1 Ratio of Specific Heats of Multi-Compound Gas Mixtures

$$\frac{1}{\gamma - 1} = \sum \frac{mol\%}{\gamma_i - 1} \quad [3] \quad (2.1)$$

2.2 Stagnation Pressure of Exhaust Gasses

$$P_s = P_1 \left(1 + \left(\frac{\gamma - 1}{2} \right) Ma_1^2 \right)^{\frac{\gamma}{\gamma - 1}} \quad [4] \quad (2.2)$$

2.3 Stagnation Temperature of Exhaust Gasses

$$T_s = T_c \left(1 + \left(\frac{\gamma - 1}{2} \right) Ma_1^2 \right) \quad [5] \quad (2.3)$$

2.4 Gas Exit Velocity

$$V_e = \sqrt{\frac{2\gamma}{\gamma - 1} RT_c \left(1 - \left(\frac{P_e}{P_c} \right)^{\frac{\gamma - 1}{\gamma}} \right)} \quad [6] \quad (2.4)$$

2.5 Exhaust Gas Density

$$\rho_x = \frac{P_x}{R_s T_x} \quad [7] \quad (2.5)$$

2.6 Exhaust Gas Mach Number

$$Ma_e = \frac{\sqrt{2} \sqrt{P_s \gamma \sqrt{\frac{P_e}{P_s}} - P_e}}{\sqrt{P_e \gamma - P_e}} \quad [7] \quad (2.6)$$

2.7 Isentropic Flow Pressure-Mach Relation

$$P_x = P_s \left(1 + \frac{(\gamma - 1)}{2} Ma_x^2 \right)^{-\frac{\gamma}{\gamma - 1}} \quad [7] \quad (2.7)$$

2.8 Speed of Sound

$$c_s = \sqrt{\gamma \frac{P_x}{\rho_x}} = \sqrt{\gamma R T_x} \quad [8] \quad (2.8)$$

3 Combustion Chamber Geometry

4 Throat Geometry

4.1 Nozzle Area Ratio

$$\epsilon = \frac{\left(\frac{2}{\gamma+1}\right)^{\frac{1}{\gamma-1}} \left(\frac{P_c}{P_e}\right)^{\frac{1}{\gamma}}}{\sqrt{\frac{\gamma+1}{\gamma-1} \left(1 - \left(\frac{P_e}{P_c}\right)^{\frac{\gamma-1}{\gamma}}\right)}} \quad [6] \quad (4.1a)$$

$$\epsilon = \frac{A_e}{A_t} \quad [7] \quad (4.1b)$$

$$\epsilon = \left(\frac{\gamma+1}{2}\right)^{-\frac{\gamma+1}{2(\gamma-1)}} \left(\frac{\left(1 + \frac{\gamma-1}{2} Ma_t^2\right)}{Ma_t}\right)^{\frac{\gamma+1}{2}} \quad [8] \quad (4.1c)$$

4.2 Throat Area

$$A_t = \dot{m} \frac{\sqrt{T_c}}{P_c} \sqrt{\frac{R}{\gamma}} \left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \quad [9] \quad (3.2.2a)$$

$$A_t = \frac{\dot{m}}{\rho_t V_t} \quad [7] \quad (3.2.2b)$$

4.3 Throat Temperature

$$T_t = T_s \left(1 + \frac{(\gamma-1)}{2} Ma_t^2\right)^{-1} \quad [7] \quad (3.2.3)$$

4.4 Throat Pressure

$$P_t = P_s \left(1 + \frac{(\gamma-1)}{2} Ma_t^2\right)^{-\frac{\gamma}{\gamma-1}} \quad [7] \quad (3.2.4)$$

5 Nozzle Geometry

Bibliography

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